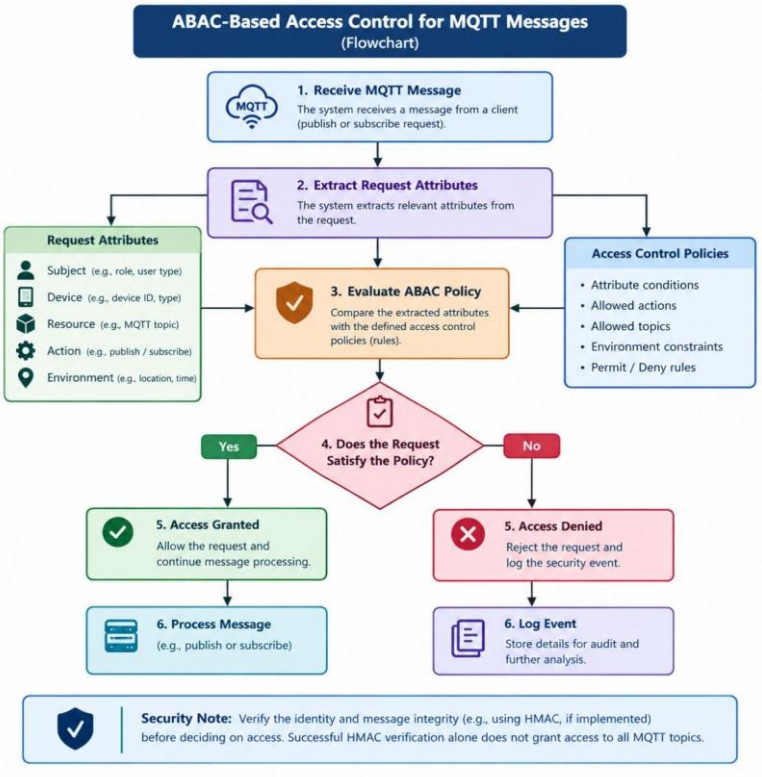
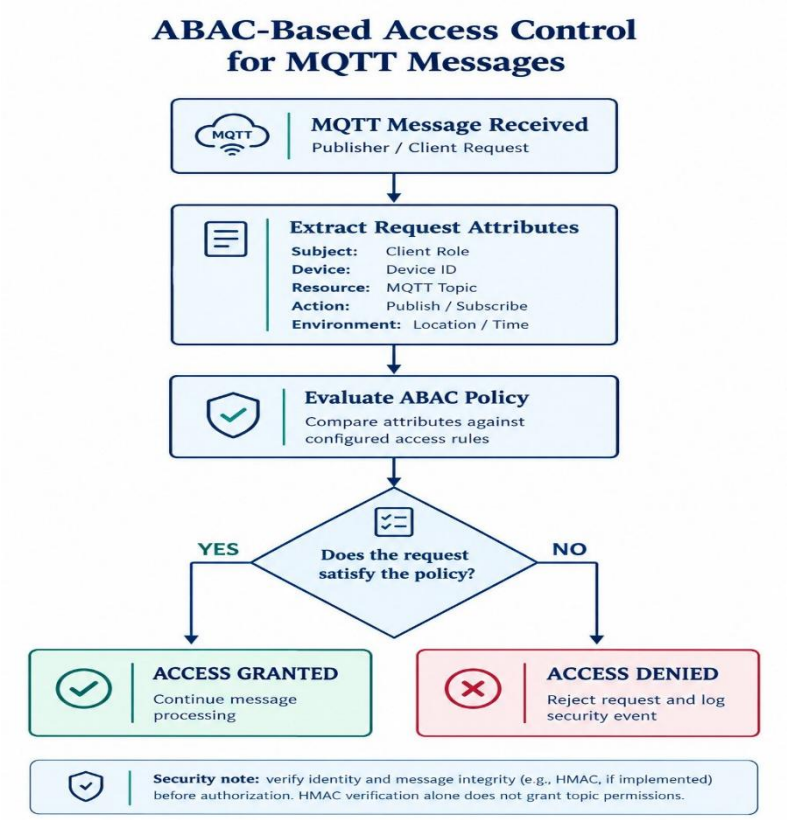
IoT Security System: Block Diagrams and Pseudocode

1. System Architecture Block Diagram :



2. ABAC-Based Access Control Flowchart for MQTT Messages :



Algorithm 1: MQTT Message Sender

BEGIN

1. Generate the sender's private and public key pair.
2. Load the receiver's public key.
3. Establish a shared secret using Elliptic Curve Diffie–Hellman (ECDH).
4. Derive the encryption key using the HMAC-based Key Derivation Function (HKDF).
5. Connect to the MQTT broker.
6. FOR each test case DO
1. Prepare the message and access-control attributes:
- Subject Role
- Device ID
- Resource
- Action
- Location

2. Serialize the message and access-control attributes into a transmittable data format.
3. Compute the HMAC of the message.
4. Encrypt the message using AES-EAX.
5. Create a packet containing:
- Sender's Public Key
- Encrypted Message
- Nonce
- Authentication Tag
- HMAC
6. IF the test case is 1 THEN
- Send a valid message with permitted access attributes.
7. ELSE IF the test case is 2 THEN
- Modify the HMAC value to simulate message tampering.
8. ELSE IF the test case is 3 THEN
- Assign an unauthorized role ("Guest").
9. ELSE IF the test case is 4 THEN
- Remove the Nonce field from the packet.
10. END IF
11. Publish the packet through MQTT.
7. END FOR
8. Close the MQTT connection.
END

Algorithm 2: MQTT Message Receiver

BEGIN
1. Generate the receiver's private and public key pair.
2. Store the receiver's public key.
3. Connect to the MQTT broker.
4. Wait for an incoming message.
5. WHEN a message is received DO
1. Parse the received packet.
2. IF the packet is missing any required fields THEN
- Print ""Access Denied"".
- Print ""Invalid Packet"".
- Terminate processing of the message.
3. END IF
4. Load the sender's public key.
5. Establish the shared secret using ECDH.
6. Derive the AES key using HKDF.
7. Decrypt the message and verify the AES-EAX authentication tag.

8. IF decryption fails or encryption integrity verification fails THEN

- Print ""Access Denied"".
- Print ""Encryption Integrity Verification Failed"".
- Terminate processing of the message.
9. END IF
10. Compute the HMAC of the received message.
11. Compare the computed HMAC with the received HMAC.
12. Extract the access-control attributes:
- Subject Role
- Device ID
- Resource
- Action
- Location
13. Apply the ABAC policy.
14. IF the access attributes do not satisfy the policy THEN
- Print ""Access Denied"".
- Print ""ABAC Policy Denied Access"".
15. ELSE IF the HMAC is invalid THEN
- Print ""Access Denied"".
- Print ""Message Tampering Detected"".
16. ELSE
- Print ""Access Granted"".
- Print ""Message Integrity Verified"".
17. END IF
6. END WHEN
END